
ZOHO INTERVIEW – QUESTION SET (L2 & L3)

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✓ LEVEL 2 (L2) QUESTIONS

1. Lucky Number in Array

Problem Statement:

You are given an integer array `arr`. A number is called a **Lucky Number** if the number itself is equal to its frequency in the array.

Task:

Return the **largest Lucky Number**.

If no Lucky Number exists, return **-1**.

Example 1:

Input:

`arr = [1, 2, 2, 3]`

Frequencies:

1 → 1 time

2 → 2 times

3 → 1 time

Lucky Numbers → 1, 2

✓ Output: 2

Example 2:

Input:

`arr = [5, 5, 5, 5]`

Frequency of 5 = 4

✗ Output: -1

2. Long Pressed Name

Problem Statement:

You are given two strings:

name → original name

typed → typed using keyboard

Sometimes characters may be **long-pressed**.

Task:

Return **true** if typed is a long-pressed version of name, else return **false**.

Example 1:

Input:

name = "alex"

typed = "aaallexx"

✓ Output: **true**

Example 2:

Input:

name = "sai"

typed = "ssaaaii"

✓ Output: **true**

Example 3:

Input:

name = "kavi"

typed = "kaaviu"

✗ Output: **false**

3. Digital Lock Minimum Moves

Problem Statement:

A digital lock has multiple dials (0–9).

Each rotation (up or down) costs **1 move**. Rotation is **circular**.

You are given:

current[] → current positions

target[] → target positions

Task:

Find the **minimum total moves** required.

Example 1:

Input:

current = [4,5,6,7]

target = [1,2,3,4]

Output:

✓ **12**

Example 2:

current = [9,0]

target = [0,9]

✓ Output: **2**

4. Sort Array by Frequency (Brute Force)

Problem Statement:

Sort the array in **descending order of frequency**.

Constraints:

No HashMap

No Collections

Stable Sort Required

Input:

arr = [6,4,7,13,4,6,4,4,13,4,6]

Frequencies:

4 → 6 times

6 → 3 times

13 → 2 times

7 → 1 time

✓ Output:

[4,4,4,4,4,4,6,6,6,13,13,7]

5. Shortest Path in Grid

Problem Statement:

You are given an $n \times m$ grid:

0 $\rightarrow$ open cell

1 $\rightarrow$ blocked cell

Start $\rightarrow$ (0,0)

End $\rightarrow$ (n-1, m-1)

You can move in **4 directions**.

Task:

Return minimum steps to reach destination.

Example Input:

```
grid = [  
  [0, 0, 0],  
  [1, 1, 0],  
  [0, 0, 0]  
]
```

✓ Output: 4

LEVEL 3 (L3) QUESTIONS

6. Custom String Encryption

Rules:

Shift alphabets by +3 (Caesar Cipher)

Digits: Replace each digit d with $(9 - d)$

Add original digits to sum n

Rotate encrypted string right by n

Reverse final string

Input:

abC124

Output:

✓ ef675d

7. Smallest Number by Concatenation

Problem Statement:

You are given an array of numbers as **strings**.

Arrange them to form the **smallest possible number**.

Input:

["3", "30", "34", "5", "9"]

✓ Output:

"3033459"

8. ATM Cash Dispensing Problem

Given:

amount to withdraw

denominations[]

stock[][]

Priority:

Always use **higher denominations first**.

Input:

amount = 2600

denominations = [2000, 500, 200, 100, 50]

stock = [[2000,1],[500,3],[200,1],[100,0],[50,100]]

✓ Output:

[2000,1]

[500,1]

[200,0]

[100,0]

[50,2]
